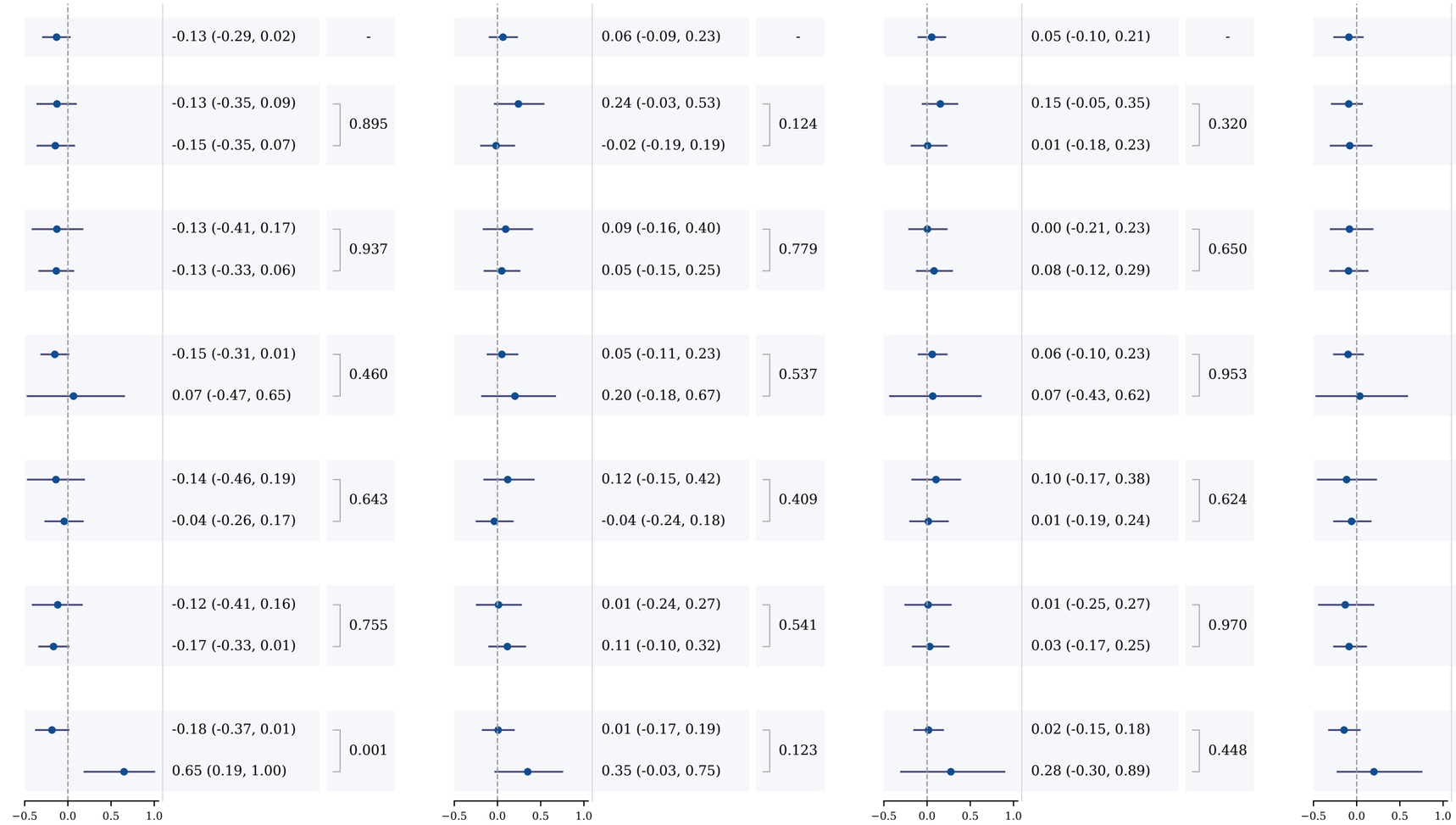
Subgroup comparison of LGBM and traditional pneumonia scores

Subgroup	N	Mortality n (%)	LGBM vs PSI		LGBM vs CURB-65		LGBM vs ADROP		LGBM vs SMART-COP	
			Delta AUROC (95% CI)	P for interaction	Delta AUROC (95% CI)	P for interaction	Delta AUROC (95% CI)	P for interaction	Delta AUROC (95% CI)	P for interaction
All	240	22 (9.2)	-0.13 (-0.29, 0.02)	-	0.06 (-0.09, 0.23)	-	0.05 (-0.10, 0.21)	-	-0.09 (-0.26, 0.07)	-
Age										
65-74 years	84	7 (8.3)	-0.13 (-0.35, 0.09)	0.895	0.24 (-0.03, 0.53)	0.124	0.15 (-0.05, 0.35)	0.320	-0.09 (-0.29, 0.06)	0.952
>=75 years	156	15 (9.6)	-0.15 (-0.35, 0.07)		-0.02 (-0.19, 0.19)		0.01 (-0.18, 0.23)		-0.08 (-0.30, 0.17)	
Sex										
Female	105	7 (6.7)	-0.13 (-0.41, 0.17)	0.937	0.09 (-0.16, 0.40)	0.779	0.00 (-0.21, 0.23)	0.650	-0.08 (-0.30, 0.18)	0.957
Male	135	15 (11.1)	-0.13 (-0.33, 0.06)		0.05 (-0.15, 0.25)		0.08 (-0.12, 0.29)		-0.09 (-0.31, 0.13)	
Immune										
Immunocompetent	209	19 (9.1)	-0.15 (-0.31, 0.01)	0.460	0.05 (-0.11, 0.23)	0.537	0.06 (-0.10, 0.23)	0.953	-0.10 (-0.26, 0.07)	0.643
Immunocompromised	31	3 (9.7)	0.07 (-0.47, 0.65)		0.20 (-0.18, 0.67)		0.07 (-0.43, 0.62)		0.04 (-0.47, 0.58)	
PSI										
I-III	157	8 (5.1)	-0.14 (-0.46, 0.19)	0.643	0.12 (-0.15, 0.42)	0.409	0.10 (-0.17, 0.38)	0.624	-0.12 (-0.45, 0.23)	0.763
IV-V	83	14 (16.9)	-0.04 (-0.26, 0.17)		-0.04 (-0.24, 0.18)		0.01 (-0.19, 0.24)		-0.06 (-0.26, 0.16)	
CURB-65										
0-1	117	10 (8.5)	-0.12 (-0.41, 0.16)	0.755	0.01 (-0.24, 0.27)	0.541	0.01 (-0.25, 0.27)	0.970	-0.13 (-0.44, 0.19)	0.811
2-5	123	12 (9.8)	-0.17 (-0.33, 0.01)		0.11 (-0.10, 0.32)		0.03 (-0.17, 0.25)		-0.09 (-0.26, 0.11)	
ADL										
>40	231	18 (7.8)	-0.18 (-0.37, 0.01)	0.001	0.01 (-0.17, 0.19)	0.123	0.02 (-0.15, 0.18)	0.448	-0.15 (-0.32, 0.03)	0.174
<=40	9	4 (44.4)	0.65 (0.19, 1.00)		0.35 (-0.03, 0.75)		0.28 (-0.30, 0.89)		0.20 (-0.22, 0.75)	



Delta AUROC = AUROC(LGBM) - AUROC(conventional score). Positive values favor LGBM.
Paired patient bootstrap 95% CIs; interaction P values are unadjusted. N is the number of patients; mortality is n (%).